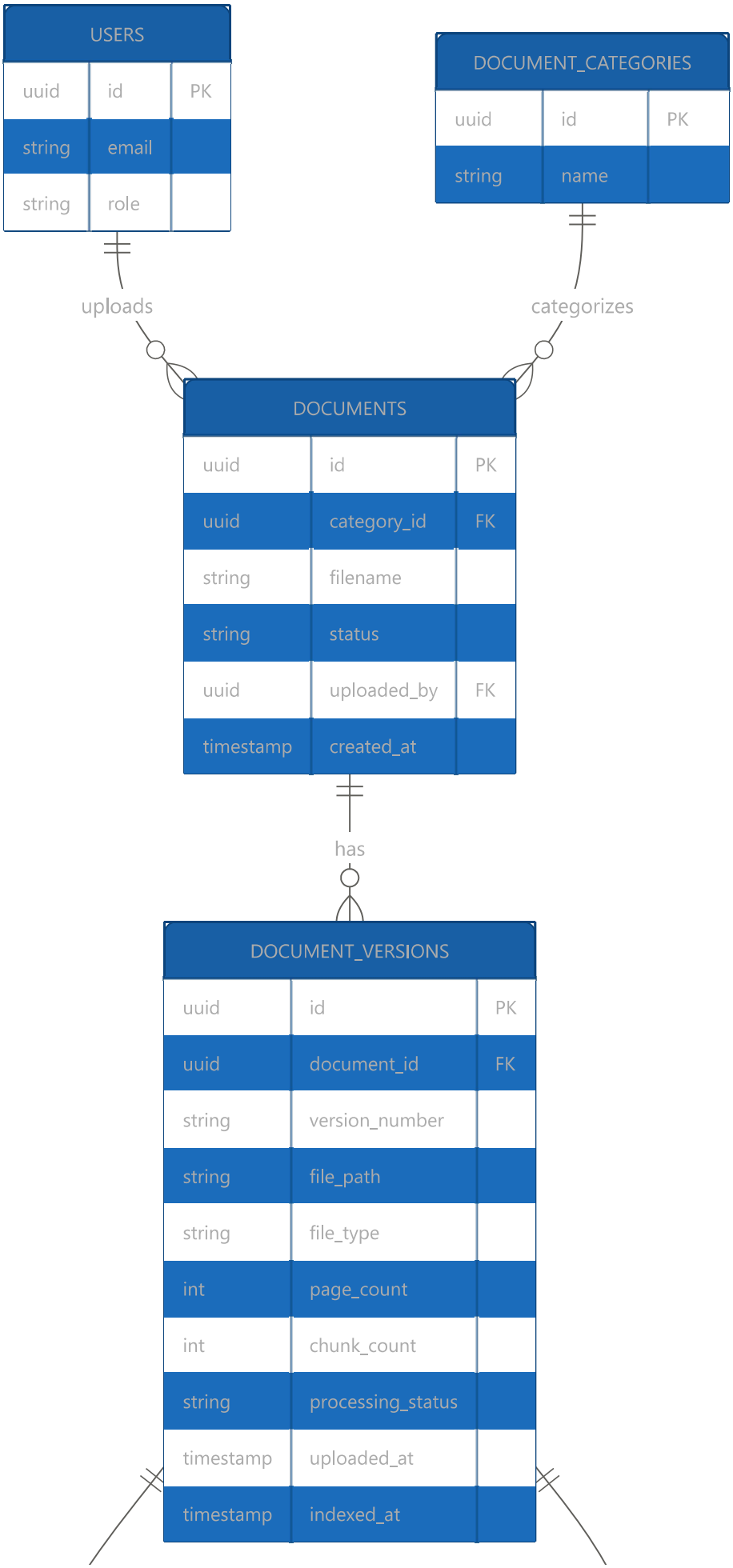
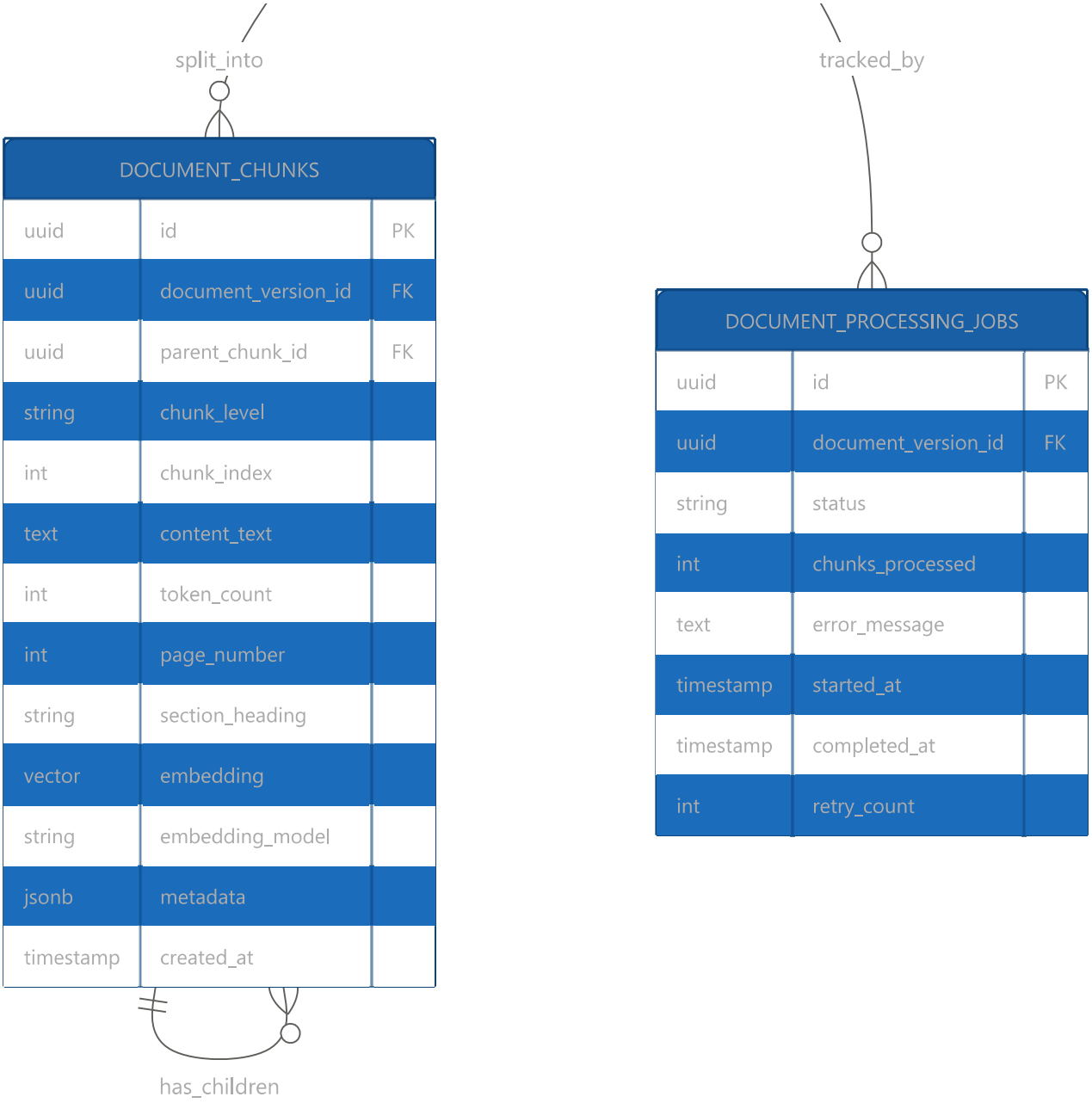
AI-Powered Internal Helpdesk Assistant — Schema

Split into three views by bounded context, instead of one dense diagram, so each stays readable on its own.

1. Document pipeline

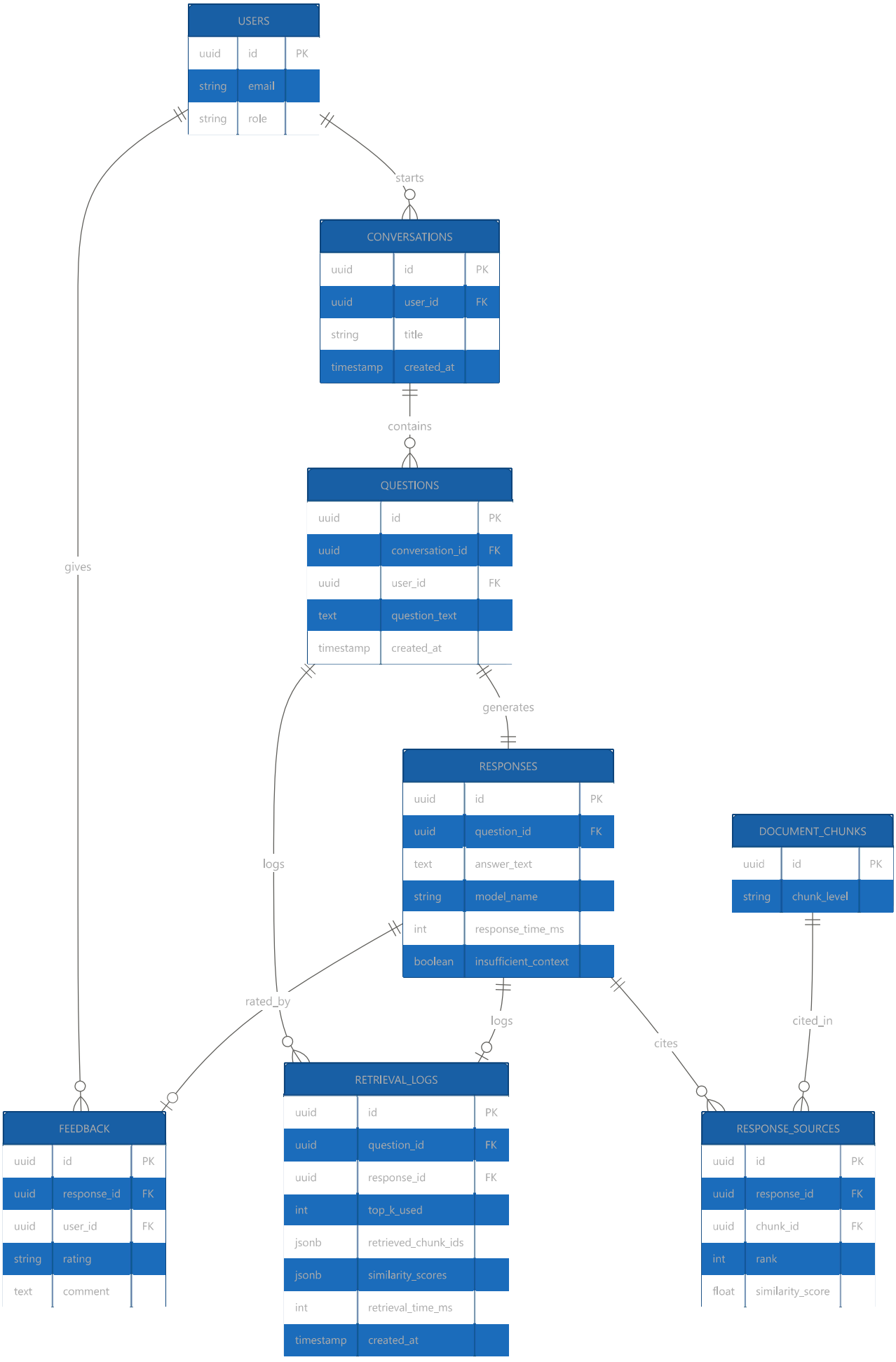
Upload → version → chunk (with parent/child splitting) → index. Tracks ingestion status per version.


USERS
uuid
id
PK
string
email
string
role
DOCUMENT_CATEGORIES
uuid
id
PK
string
name
uploads
categorizes
DOCUMENTS
uuid
id
PK
uuid
category_id
FK
string
filename
string
status
uuid
uploaded_by
FK
timestamp
created_at
has
DOCUMENT_VERSIONS
uuid
id
PK
uuid
document_id
FK
string
version_number
string
file_path
string
file_type
int
page_count
int
chunk_count
string
processing_status
timestamp
uploaded_at
timestamp
indexed_at



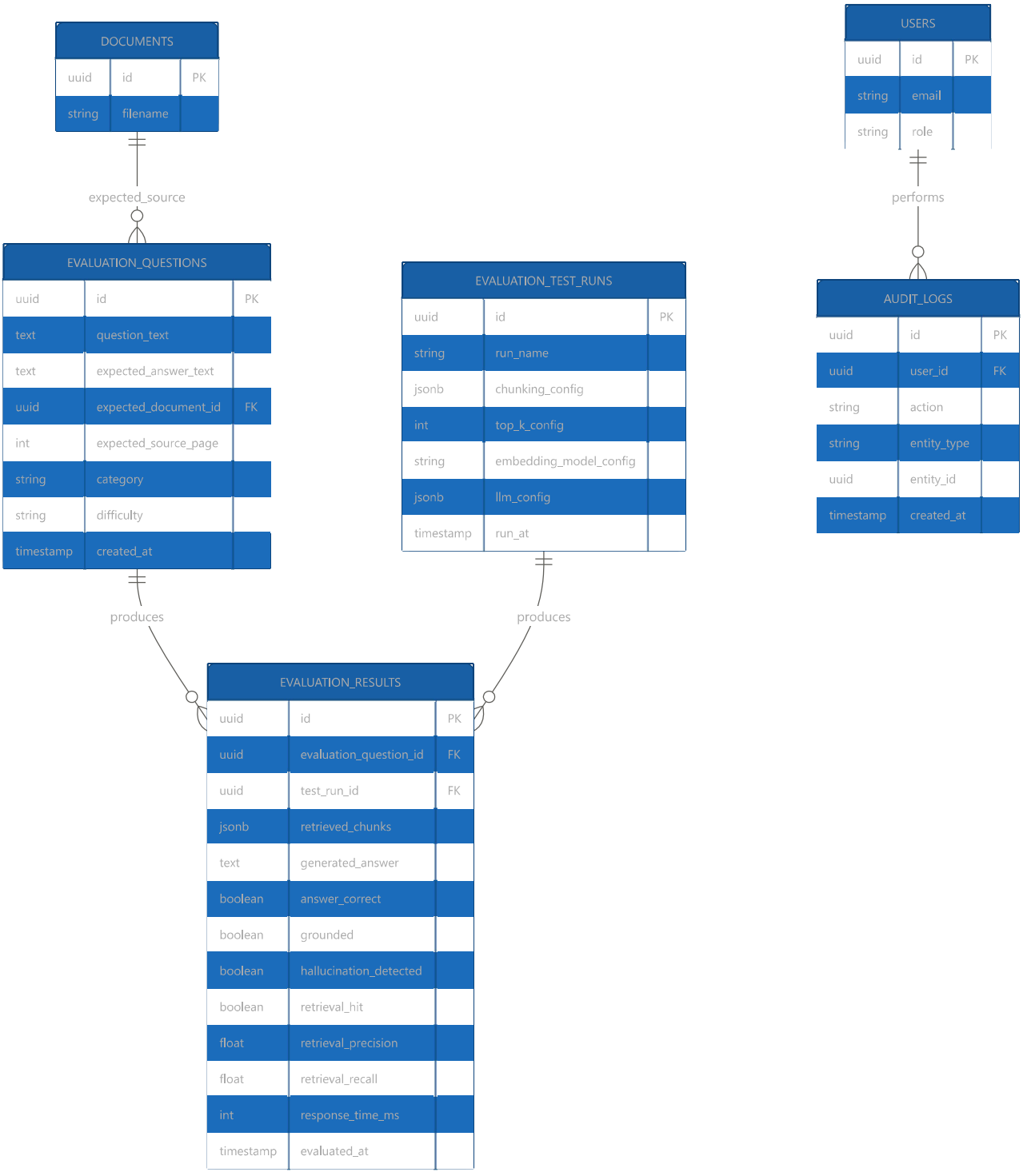
2. Conversation & retrieval

Employee questions, generated answers, the chunks cited as sources, feedback, and raw retrieval logs.


USERS
uuid
id
PK
string
email
string
role
starts
CONVERSATIONS
uuid
id
PK
uuid
user_id
FK
string
title
timestamp
created_at
contains
QUESTIONS
uuid
id
PK
uuid
conversation_id
FK
uuid
user_id
FK
text
question_text
timestamp
created_at
gives
generates
logs
RESPONSES
uuid
id
PK
uuid
question_id
FK
text
answer_text
string
model_name
int
response_time_ms
boolean
insufficient_context
DOCUMENT_CHUNKS
uuid
id
PK
string
chunk_level
rated_by
logs
cites
cited_in
FEEDBACK
uuid
id
PK
uuid
response_id
FK
uuid
user_id
FK
string
rating
text
comment
RETRIEVAL_LOGS
uuid
id
PK
uuid
question_id
FK
uuid
response_id
FK
int
top_k_used
jsonb
retrieved_chunk_ids
jsonb
similarity_scores
int
retrieval_time_ms
timestamp
created_at
RESPONSE_SOURCES
uuid
id
PK
uuid
response_id
FK
uuid
chunk_id
FK
int
rank
float
similarity_score

3. Evaluation & governance

The offline test set used to score retrieval/answer quality across experiment runs, plus the audit trail.



Note: USERS, DOCUMENTS, and DOCUMENT_CHUNKS appear as trimmed reference stubs in more than one diagram above — their full field list lives in the diagram where they're the primary subject (USERS and DOCUMENTS in Document

pipeline; DOCUMENT_CHUNKS also in Document pipeline).